

Exact low-dimensional numerical-sparsity uncertainty constants and a stronger asymptotic lower bound

Candidate partial solution packet for arXiv:1504.01014, Problem 3

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Status. Candidate substantial partial solution, likely valid. We determine $c(1), c(2), c(3)$ exactly and improve the source's asymptotic lower bound by a factor of 16.2. The exact problem for $n \geq 4$ remains open.

1 Source problem

For $x \in \mathbb{C}^n \setminus \{0\}$, Bandeira, Lewis, and Mixon define the *numerical sparsity*

$$\text{ns}(x) = \frac{\|x\|_1^2}{\|x\|_2^2}.$$

Problem 3 on page 5 of [1] asks for the exact value of

$$c(n) = \frac{1}{n} \sup_{U \in \text{U}(n)} \inf_{x \neq 0} (\text{ns}(x) + \text{ns}(Ux)). \quad (1)$$

The source proves $\liminf c(n) \geq 1/540000$ and records the upper bound $c(n) < 1 + 1/n$ for $n \geq 4$.

our uncertainty principle establishes that the Fourier transform is rare in that the vast majority of unitary matrices offer much more uncertainty in the worst case. This naturally leads to the following question:

Problem 3. For each n , what is the largest $c = c(n)$ for which there exists a unitary matrix U that satisfies $\text{ns}(x) + \text{ns}(Ux) \geq cn$ for every $x \in \mathbb{C}^n \setminus \{0\}$?

Letting $x = e_1$ gives $\text{ns}(x) + \text{ns}(Ux) \leq 1 + \|Ux\|_0 \leq n + 1$, and so $c(n) \leq 1 + o(1)$; a bit more work produces a strict inequality $c(n) < 1 + 1/n$ for $n \geq 4$. Also, our proof of the uncertainty principle implies $\liminf_{n \rightarrow \infty} c(n) \geq 1/540000$.

Figure 1: Problem 3 and the original bounds on page 5 of [1].

2 Proof intuition

In dimensions two and three, the trivial basis-vector upper bound can be attained by the Fourier matrix. For a unit vector z , numerical sparsity is one plus twice the sum of the magnitudes of all pairwise coordinate products. For F_2 , the sole output pair product is $(x_0^2 - x_1^2)/2$. For F_3 , three phase-weighted sums of the three output pair products isolate, up to unit phases, the quadrics $x_0^2 - x_1x_2$, $x_1^2 - x_0x_2$, and $x_2^2 - x_0x_1$. Choosing the largest coordinate then turns the desired estimate into the elementary inequality $a(a + b + c) \geq a^2 + b^2 + c^2$.

For the asymptotic bound, we keep the source's random-unitary restricted isometry theorem but replace its coarse width lemma. Sorting a null vector into decreasing blocks of size k controls the head by RIP and the tail by the ℓ_1 norm. Pythagoras then gives a width constant valid for every $\delta < 1$, allowing a much larger RIP distortion and a much larger k .

3 The first three constants

Let F_n denote the normalized Fourier matrix

$$(F_n x)_j = \frac{1}{\sqrt{n}} \sum_{r=0}^{n-1} \omega_n^{jr} x_r, \quad \omega_n = e^{-2\pi i/n}.$$

Theorem 1 (Exact low-dimensional values). *For the constants in (1),*

$$\boxed{c(1) = 2, \quad c(2) = \frac{3}{2}, \quad c(3) = \frac{4}{3}.}$$

The matrices F_2 and F_3 attain the two nontrivial suprema.

Proof. For every unit vector $v \in \mathbb{C}^n$, Cauchy–Schwarz gives $\text{ns}(v) \leq n$. Testing an arbitrary U at a coordinate vector therefore gives

$$\inf_{x \neq 0} (\text{ns}(x) + \text{ns}(Ux)) \leq n + 1. \quad (2)$$

This proves the required upper bounds 3 and 4 in dimensions two and three. The case $n = 1$ is immediate.

For $n = 2$, normalize $\|x\|_2 = 1$ and put $y = F_2 x$. Then

$$\text{ns}(x) + \text{ns}(y) = 2 + 2(|x_0 x_1| + |y_0 y_1|), \quad y_0 y_1 = \frac{x_0^2 - x_1^2}{2}.$$

Write $a = \max(|x_0|, |x_1|)$ and $b = \min(|x_0|, |x_1|)$. The reverse triangle inequality gives

$$\begin{aligned} |x_0 x_1| + |y_0 y_1| &\geq ab + \frac{a^2 - b^2}{2} \\ &\geq \frac{a^2 + b^2}{2} = \frac{1}{2}, \end{aligned}$$

where the second inequality is $ab \geq b^2$. Thus the uncertainty sum is at least 3, and equality holds at a coordinate vector. Hence $c(2) = 3/2$.

For $n = 3$, define

$$S(z) = |z_0 z_1| + |z_1 z_2| + |z_2 z_0|.$$

For unit z one has $\text{ns}(z) = 1 + 2S(z)$. Put $y = F_3 x$ and $\omega = \omega_3$. Direct expansion, using $1 + \omega + \omega^2 = 0$, gives

$$y_0 y_1 + y_1 y_2 + y_2 y_0 = x_0^2 - x_1 x_2, \quad (3)$$

$$y_0 y_1 + \omega y_1 y_2 + \omega^2 y_2 y_0 = \omega(x_1^2 - x_0 x_2), \quad (4)$$

$$y_0 y_1 + \omega^2 y_1 y_2 + \omega y_2 y_0 = \omega^2(x_2^2 - x_0 x_1). \quad (5)$$

Each left side has modulus at most $S(y)$. Relabel the coordinates so that their magnitudes are $a \geq b \geq c$. The identity corresponding to the largest coordinate and the reverse triangle inequality yield

$$S(y) \geq a^2 - bc.$$

Consequently,

$$S(x) + S(y) \geq ab + ac + bc + a^2 - bc = a(a + b + c) \geq a^2 + b^2 + c^2 = 1.$$

It follows that

$$\text{ns}(x) + \text{ns}(F_3 x) = 2 + 2(S(x) + S(F_3 x)) \geq 4.$$

Again a coordinate vector gives equality. Combining with (2) proves $c(3) = 4/3$. $\square$

4 A direct RIP-to-width estimate

We use the standard definition that a matrix Φ satisfies (k, δ) -RIP if

$$(1 - \delta)\|w\|_2^2 \leq \|\Phi w\|_2^2 \leq (1 + \delta)\|w\|_2^2$$

for every k -sparse w .

Lemma 2 (Nullspace width for every $\delta < 1$). *If Φ satisfies (k, δ) -RIP with $0 < \delta < 1$, then every $z \in \ker \Phi$ satisfies*

$$\|z\|_2^2 \leq \frac{2}{(1 - \delta)k} \|z\|_1^2. \quad (6)$$

Proof. Partition the coordinates of z into blocks $S_0, S_1, \dots$ of size k (except possibly the last), in decreasing order of $|z_i|$. For $j \geq 1$,

$$\|z_{S_j}\|_2 \leq \frac{1}{\sqrt{k}} \|z_{S_{j-1}}\|_1,$$

and hence, with $A = \|z\|_1 / \sqrt{k}$,

$$\sum_{j \geq 1} \|z_{S_j}\|_2 \leq A, \quad \|z_{S_0^c}\|_2 \leq A. \quad (7)$$

Since $\Phi z = 0$, RIP and the triangle inequality give

$$\sqrt{1 - \delta} \|z_{S_0}\|_2 \leq \|\Phi z_{S_0}\|_2 \leq \sqrt{1 + \delta} \sum_{j \geq 1} \|z_{S_j}\|_2 \leq \sqrt{1 + \delta} A.$$

Using the orthogonality of the two coordinate pieces and (7),

$$\|z\|_2^2 = \|z_{S_0}\|_2^2 + \|z_{S_0^c}\|_2^2 \leq \left(\frac{1 + \delta}{1 - \delta} + 1 \right) A^2 = \frac{2}{1 - \delta} \frac{\|z\|_1^2}{k}.$$

□

5 Improved asymptotic constant

We use Theorem 8 of [1]: if U is Haar unitary, then $\Phi = [I \ U]$ satisfies (k, δ) -RIP with positive probability for all sufficiently large n provided $\delta < 1$ and

$$n \geq \frac{256}{\delta^2} k \log \left(\frac{en}{k} \right). \quad (8)$$

Theorem 3 (Improved asymptotic lower bound). *For every sufficiently large n , there exists $U \in \mathbf{U}(n)$ such that*

$$\text{ns}(x) + \text{ns}(Ux) \geq \frac{3}{20} \left\lfloor \frac{n}{5000} \right\rfloor \quad (x \in \mathbb{C}^n \setminus \{0\}).$$

Consequently,

$$\boxed{\liminf_{n \rightarrow \infty} c(n) \geq \frac{3}{100000}}.$$

Proof. Take $\delta = 7/10$ and $k = \lfloor n/5000 \rfloor$. The function $\alpha \mapsto \alpha \log(e/\alpha)$ is increasing on $(0, 1)$. Therefore

$$\begin{aligned} \frac{256}{\delta^2} \frac{k}{n} \log\left(\frac{en}{k}\right) &\leq \frac{25600}{49} \frac{1}{5000} \log(5000e) \\ &= \frac{256}{2450} \log(5000e) < \frac{256}{2450} (9.52) < 1. \end{aligned}$$

Thus (8) holds. The probability in [1, Theorem 8] is positive once n is sufficiently large, so there exists a unitary U for which $\Phi = [I \ U]$ has this RIP.

For nonzero x , put $z = (Ux, -x) \in \ker[I \ U]$. Lemma 2 gives

$$\frac{\|z\|_1^2}{\|z\|_2^2} \geq \frac{(1-\delta)k}{2} = \frac{3k}{20}.$$

On the other hand, unitarity and the elementary inequality $(a+b)^2/2 \leq a^2 + b^2$ give

$$\begin{aligned} \frac{\|z\|_1^2}{\|z\|_2^2} &= \frac{(\|x\|_1 + \|Ux\|_1)^2}{2\|x\|_2^2} \\ &\leq \frac{\|x\|_1^2 + \|Ux\|_1^2}{\|x\|_2^2} = \text{ns}(x) + \text{ns}(Ux). \end{aligned}$$

This proves the finite- n assertion. Divide by n and let n tend to infinity. □

6 Limitations and novelty check

Theorem 1 does not determine $c(4)$ or any later exact value. Theorem 3 improves only the lower constant; it does not identify the asymptotically optimal one. Numerical exploration of F_4 , F_5 , and a small sample of Haar unitaries was used only to decide where further exact arguments cease to be credible and supplies no proof claim.

A bounded literature search on 11 August 2026 used the exact source title and arXiv id, the authors, the formula $\text{ns}(x) + \text{ns}(Ux)$, “numerical sparsity optimal unitary,” and small-dimensional discrete Fourier ℓ_1 uncertainty. No later paper explicitly answering Problem 3 or recording the values $c(2), c(3)$ was found. Nearby finite-dimensional uncertainty papers use different localization functionals. Novelty confidence is moderate for the exact small-dimensional values and moderate-to-high for the explicit factor-16.2 improvement obtained from the source’s own RIP theorem.

Human-review recommendation. Check the Fourier phase convention in (3)–(5), the largest-coordinate inequality in dimension three, the decreasing-block bound (7), and the numerical inequality verifying (8). If these pass, retain this as a substantial partial answer to Problem 3.

References

- [1] A. S. Bandeira, M. E. Lewis, and D. G. Mixon, *Discrete uncertainty principles and sparse signal processing*, J. Fourier Anal. Appl. 24 (2018), 935–956; arXiv:1504.01014.